Application: Redefining amino acid requirements in pregnancy for precision sow feeding

Crystal Levesque - crystal.levesque@sdstate.edu
2016 New Innovator in Food and Agriculture Research Award

Summary

ID: 0000000004
Status: Final Reporting
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Final Scientific Progress Report

Completed - Jan 21 2022

Form for "FFAR Annual Progress Report"

Grant Information*

Award Program	2016 New Innovator in Food and Agriculture Research Award
Grant ID	0000000004
Project Title	Redefining amino acid requirements in pregnancy for precision sow feeding
Grant Start Date	12/31/2016
Grant End Date	12/30/2021
Total Budget	$599,979.00
Matching Funder(s)	ADM Animal Nutrition
Final Report Date	19 January 2022

Project Director/Principal Investigator Information*

Full Name	Dr. Crystal Levesque
Email	crystal.levesque@sdstate.edu
Phone	605-688-5011

Grantee Organization Information

Organization Name	South Dakota State University
Address	Box 2201, SAD 200
City	Brookings
State	South Dakota
ZIP	57007
EIN	46-6000364

Authorized Signing Official Information

Name	Dr. Dianne Nagy
Email	dianne.nagy@sdstate.edu
Phone	605-688-5051

1. GENERAL INFORMATION

Notes on Spending

If you've experienced any deviation in the spending for your award for the last reporting period such as changes to the amount of source or matching funds, changes from your original spending plan, or carry-forward, please provide a justification below. You may enter "N/A" or "Nothing to report" if you did not experience any special circumstances.

N/A

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.*

Brookings, SD, At large

1.2. How many new jobs were created by the grant during this reporting period? *

Jobs maintained by the grant during this reporting period relate to graduate students and post-doctoral fellows. See questions related to educational training below.

1.3. How many jobs were maintained by the grant during this reporting period?*

Jobs maintained by the grant during this reporting period relate to graduate students and post-doctoral fellows. See questions related to educational training below.

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant?*

No

If yes, please explain:

N/A

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.*

No

2. SCIENTIFIC REPORT

2.1. Public Abstract*

The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Application of precision feeding requires mathematical models and technology to adjust daily feeding. For the pregnant sow, technology exists for precision feeding; however, the available nutrient requirement models are based on limited data. The reproductive sow is the cornerstone of pork production and she contributes substantially to manure nutrient excretion. In a similar way, dietary amino acid (AA) supply is the foundation of protein deposition. This project is focused on biological processes that influence AA requirements during pregnancy, redefining methods to establish AA requirements, and determining AA requirements at key periods during pregnancy.
Fundamental to modelling AA requirements are the loses of dietary AA because not all AA are used for processes that result in gain of protein tissue; inefficiency increases, AA requirement. We determined that dietary lysine (considered the primary AA for protein tissue gain) was used with similar efficiency in the first and third trimester, but more efficiently in the second. However, dietary threonine was more efficiently used in the second and third trimester and least efficiently in the first. This is important because it indicates that the metabolic processes driving AA requirements in each stage of pregnancy change. No constant 'efficiency coefficient' for all AA and all stages of gestation is suitable.
Whole body nitrogen retention is primarily used to define AA requirements for pigs because growth in muscle tissue is highly correlated to retention of nitrogen. In fact, the point of maximal nitrogen retention is considered the AA requirement. However, we determined that maximal nitrogen retention may not

suitably reflect the AA necessary to optimize gain in maternal muscle mass and fetal tissue development. In late pregnancy, maximal nitrogen retention reflects approximately 70% of the total AA requirement. While nitrogen retention is relatively easily measured, markers of fetal tissue gain are not. Using the adjustment factor of 70% allows determination of true AA requirements that account for fetal tissue gain using well established research methods.

We combined data available in the literature with data generated from this project to develop a new model describing AA gain in pregnancy-related tissues across stage of pregnancy. We learned that in late pregnancy leucine supply, rather than lysine, may be the determining factor for maternal tissue mobilization which occurs when dietary AA supply is insufficient to meet AA demand. We also learned that the profile, and quantity, of AA needs change as pregnancy progresses. This is important because at present in commercial swine production, the same diet (i.e. constant AA profile) is fed throughout pregnancy. The single diet strategy, adjusting only for daily quantity, is not suitable to meet the changing AA needs of the pregnant sow. As a result, efficiency of diet AA use is reduced, maternal tissue mobilization is increased, and potentially more nitrogen is lost to the environment.

The fundamental data generated by this program advances precision feeding of sows contributing to the production of a globally sustainable, high quality protein source for humans while ensuring a food supply with strong economic and environmental resilience.

2.2 Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making.*

The findings can be used in two key ways. 1) guiding future planning or decision-making: the AA requirement model developed in the final phase of this project has been made into an interactive web-based model where users can alter model parameters such as female age, day of gestation, expected litter size to generate predicted AA requirements. For the research community this interactive model allows development of strong empirical studies to test model predictions and evaluate the impact of additional parameters not currently included, but relevant, to more advanced precision feeding models such as supply of other nutrients (i.e. energy), environmental temperature, housing conditions, and genetics. 2) practical application by the agriculture community: our work confirmed previous recommendations that a single diet strategy in gestation was not sufficient to meet all AA demands, particularly in late pregnancy. While AA requirement predictions of the new model must still be challenged and validated by well designed studies, this work supports implementation of feeding at least two different diets formulated to achieve different AA profiles during gestation under commercial pork production conditions.

2.3. Research Outcomes/Impact *

The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- Potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- Intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- End outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes
- One of the primary potential outcomes of this work is the confirmation that the “single diet” strategy (i.e. providing a single diet and thus a constant profile of AA through the entirety of gestation), adjusting only for daily quantity, is not suitable to meet the changing AA needs of the pregnant sow. The practice of a single gestation diet means efficiency of diet AA use is reduced, maternal tissue mobilization is increased, and potentially more nitrogen is lost to the environment. A shift to using a phased approach (i.e. phase feeding) to gestating sow feeding where a different diet was provided in the third trimester would improve efficiency of production (both nutritionally and environmentally).
- while phase feeding has potential to improve production efficiency, advanced precision feeding of sows requires testing and validation of the new model developed from this work. For example, empirical AA requirement studies evaluating traditional response variables (i.e. nitrogen balance and retention) in combination with metabolic function variables confirming that leucine supply in late gestation drives maternal tissue mobilization. These studies will provide the basis for stronger requirement models that are more predictive across a range of conditions (i.e. sow age, environment, litter size) that are ultimately required to establish optimal precision feeding of the pregnant sow.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets.*

The key avenues of inquiry that were proposed to be addressed by this research program were:
1) Identification of underlying biological processes (i.e. AA utilization efficiency) that influence AA requirements during pregnancy. 100% complete
2) Defining methods to accurately determine AA requirements in pregnant swine. 100% complete
3) Determine AA requirements at key periods during pregnancy. 70% complete

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award?*

Yes

If so, please list the goal(s) that have changed and provide justification for the change from the approved goals.*

In year 4, the focus of investigation during this period was centered on model development and practical formulation. The original award focused on specific studies to enhance the most recent available model at the time of grant submission (National Research Council 2012). Results of the first 3 years of this project demonstrated a need to develop a revised model based on our work and more recently published data. While this is a change in the specific goal, the overall goal and impact of the award on sustainability of pork production has not changed.

2.6. What was accomplished under the goals/specific aims or milestones for the project?*

Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include a discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like.

Accomplishments over the course of the project

Major activities: The major activities were live animal studies assessing aspects of AA metabolism during pregnancy and practical feeding trial and development of AA deposition during pregnancy model. A total of 4 AA metabolism studies using pregnant sows and pubertal gilts and 1 practical precision feeding in gestation trial were conducted. This data and data from published literature were used for model development.

The major activities were accomplished to meet the following specific objectives. Objectives are not listed in chronological order:

1. Determine the impact of stage of gestation and amino acid on efficiency of dietary AA utilization.
2. Model the pattern of AA deposition in maternal tissue and products of conception during pregnancy and as influenced by digestible AA intake
3. Test developed model against published empirical data sets
4. Assess impact of phase feeding in gestation on sow reproductive performance, offspring robustness and offspring growth to market over 2 successive parities
5. Determine the effect of graded methionine intake in pubertal gilts on nitrogen retention and plasma AA concentrations
6. Determine the impact of dietary methionine level during mid and late gestation in older parity sows on reproductive performance.

Significant results for each objective listed above in order:

1. Efficiency of dietary AA utilization during pregnancy: 1) efficiency of use (kSID) of lysine and threonine are not constant across gestation as assumed in current AA requirement models, 2) kSID of threonine in early gestation appears to be lesser than lysine but similar to lysine in late gestation suggesting specific tissue driving growth in each stage of pregnancy influences efficiency of use and differentially alters AA requirements.
2. Model development: 1) what was previously loosely defined as ‘time-dependent’ protein deposition is better described as gain in pregnancy-associated fluids and tissues coincident with lower maternal protein deposition between d40 and 70 of gestation in combination with decreased protein deposition in early gestation relative to pre-breeding levels. 2) pregnancy-related tissues increase in weight but also protein density as gestation progresses where previous models assumed a constant protein content, 3) quantification of negative maternal protein deposition as influenced by litter size and parity in late gestation, 4) leucine demand for pregnancy-related tissues may drive the mobilization of maternal tissue to meet conceptus demands rather than lysine as previously assumed, 4) the AA profile of the protein deposited changes throughout pregnancy indicating that a constant dietary AA ratio is not suitable to

meet sow and fetal demands, particularly in late gestation.

3. Substantiating new model with empirical data sets: 1) a linear-logistic model better describes the pattern of nitrogen retention to increasing digestible AA intake than linear-plateau model most commonly used where nitrogen retention peaks, declines, then increases again, 2) the point of lower nitrogen retention was associated with greater total piglets born suggesting AA intake at peak nitrogen retention may not sufficiently reflect optimal AA intake/requirement and may not account for optimal AA demand for metabolic functions beyond lean tissue accretion, 3) AA intake needed to meet optimal AA demand for maternal and fetal tissue is estimated to be Nretmax/0.70 where Nretmax = maximal nitrogen retention.

4. Practical application of precision feeding in gestation: 1) precision feeding had minor impacts on sow body weight or litter characteristics at birth over two successive parities, 2) precision feeding in gestation resulted in heavier pigs at weaning, 3) the weight difference at weaning was maintained through to market resulting in up to 4 days fewer on feed

5. Methionine requirement for lean tissue and metabolic demand: 1) nitrogen retention was minimized at methionine intake that maximized plasma taurine and methionine concentration supporting the idea that variables reflective of metabolic status and fetal development should be used to establish pregnancy-specific AA requirements in combination with nitrogen balance.

6. Methionine level in mid and late gestation: as noted in section 2.10, animal work required to accomplish this objective was stopped due to a disease outbreak at the sow farm.

Key outcomes:

- AA requirements based on maximal nitrogen retention alone does not adequately reflect AA demands to meet both maternal and fetal needs during pregnancy. Empirical studies considering both lean tissue demand and metabolic status are ultimately required to reestablish the essential AA requirements during pregnancy in the sow. However, these studies can be very difficult to accomplish; the adjustment factor Nretmax/0.70 maybe a partial solution to establishing new AA gestational requirements.
- The single diet strategy, adjusting only for daily quantity, is not suitable to meet the changing AA needs of the pregnant sow and developing fetus'. More closely meeting all demands for AA results in more robust offspring with the potential for greater lifetime gain.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.*

- live animal protocols were reviewed by at least 1 other scientist with relevant expertise and vetted through the Institutional Animal Care and Use Committee
- experimental diets were labelled with non-descript but clearly differentiable labels (i.e. treatment “A”, “B”, “C”) to reduce bias during data collection, review and analysis.
- review of data analysis by expert in statistics

2.8. List key stakeholders that could be served by results of this project.*

- commercial and consulting swine nutritionists
- policy makers
- commercial swine production operations of all sizes and types (i.e. organic, antibiotic-free production, conventional, pasture-raised)

2.9. How have the results of this reporting period been disseminated to communities of interest?*

Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products.

Dissemination of results and discussion of their potential implications for swine production have centered around public presentation of results at scientific conferences. Details of these presentations are provided in Information Products.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them.*

Only describe significant challenges that may have impeded the research and emphasize their resolution.

The COVID-19 pandemic continued to delay progress on starting the last graduate student on this project. After returning to regular work schedules, plans were put in place for two additional trials. Shortly after starting these trials the swine barn broke with the highly contagious Porcine Respiratory and Reproductive Syndrome (PRRS). Because of the potential impact of the virus that causes this disease and the consequences for research and education activities, the decision was made to depopulate the swine farm. That depopulation/repopulation plan is currently underway; however, that decision ended any further work on this project.

2.11. What opportunities for training and professional development has the project provided during this reporting period?*

If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals.

Work conducted this past year is the basis of a PhD program for one student. This student is expected to defend in summer 2022. This student was recently awarded an upcoming investigator award by the Midwest section of the American Society of Animal Science. The work conducted by this student significantly contributed to his success in receiving this award.
The post-doctoral researcher involved in this project successfully secured a Research Scientist position at South Dakota State University and is currently developing their own line of research investigating functional AA requirements in nursery pigs.
One visiting scholar was added to the project in this reporting period.
Routine weekly meetings were held between the PI, students and post-doctoral participant to discuss research data, data interpretation, discussion of challenges and opportunities to overcome/address challenges and develop final steps. The meetings provided opportunity to further develop critical thinking skills. As part of the professional development training activities, the graduate student participated in a communication training program focused on learning to communicate science to non-scientists. The student completed a 2-day course focused on how to transfer scientific knowledge to media, consumers, government, and livestock producers. The 2-day training concluded with formal presentations where each participating student presented research to a targeted non-scientist audience to a panel of relevant experts and received direct feedback on the quality and success of the presentation. The PhD student was also provided opportunity to disseminate research to the scientific community as outlined in section 2.9.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail.*

PhD student: 1; post-doctoral researcher: 1; undergraduate visiting scholar: 1

3. INFORMATION PRODUCTS

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.*

conference abstracts: 2

3.2. Please provide a list of citations for the information products produced during this period.*

Ramirez-Camba, C.R. and Levesque, C.L. An empirical model of essential amino acid requirements in the pregnant sow. Midwest ASAS Annual Meeting, March 14 – 16, 2022

Ramirez-Camba, C.R. and Levesque, C.L. A mechanistic model of growth and amino acid deposition in the pregnant sow: model development, evaluation, and application. Midwest ASAS Annual Meeting, March 1 – 3, 2021.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award?*

No

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.*

https://sdsuswine.shinyapps.io/sows/ Web-based interactive program of the AA deposition model developed. Users can alter sow parameters such as age, body weight, expected litter size and litter weight to model how these parameters alter AA deposition in maternal tissue and products of conception.

3.5. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period?*

No

3.6. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements?*

Yes

If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely.

The algorithms used to describe the changes in AA deposition for specific tissues are not available for public review at this point. An invention disclosure form regarding model algorithms was submitted in December 2021.

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?*

Principal Investigator contact information is provided in all public dissemination of results/activities. Raw data from empirical studies and access to the web-based model program are made available upon direct contact with PI.

4. DATA MANAGEMENT

4.1. During this period, did the project generate any data?*

Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries).

Yes

Please list the data generated in this reporting period.

Data sets related to gilt response to graded levels of dietary methionine were generated during this reporting period. These data sets include:

- Gilt's body weight over the 60d experimental period (164 – 212 days of age)
- Nitrogen input (i.e. diet nitrogen) and output (urinary and fecal nitrogen) during the 5-d nitrogen balance collections
- Calculations of total tract nitrogen digestibility and whole body retention
- Serum AA concentrations prior to and at completion of nitrogen balance collections

Data sets related to compilation of AA efficiency data and published data of sow response to dietary AA levels.

4.2. If you list multiple data sets, are these data sets related?*

Yes

If so, please provide a short description of how they are related.*

The data sets on nitrogen balance (input and output), along with serum AA concentrations were used to determine the maternal methionine requirement during pregnancy. The compiled data sets were used to develop the AA deposition model.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.*

FFAR is developing a more nuanced Data Management Policy. As part of this effort, we are introducing new questions into the Annual Reports on Data. All questions on this page are required to have a response. In cases where your answer may be repeated, you are welcome to copy-paste relevant information. Thank you for your patience as we update our reporting.

Are the data experimental, observational, simulation, or compiled/derived? Check all categories that apply.

Responses Selected:

Experimental: Data captured through a controlled experiment where an intervention is being tested. Results should be replicable and reproducible (e.g., lab experiments, field trials)
Compiled data: Data compiled using extant data sources and refined to develop a meta-analyses or systems reviews

For each checked box, please provide a brief description of the data including where the data are collected / sourced.

Experimental: Data sets related to gilt response to graded levels of dietary methionine were generated during this reporting period. These data sets include:

- Gilt's body weight over the 60d experimental period (164 – 212 days of age)
- Nitrogen input (i.e. diet nitrogen) and output (urinary and fecal nitrogen) during the 5-d nitrogen balance collections
- Calculations of total tract nitrogen digestibility and whole body retention
- Serum AA concentrations prior to and at completion of nitrogen balance collections

Compiled: Data sets related to compilation of AA efficiency data and published data of sow response to dietary AA levels

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Describe (what, where, when) and list additional data (not captured in question above), plus samples, physical collections, software, curriculum materials and other materials produced during this period. Who is the expert or person(s) tasked with data collection (please provide their name and affiliation with the project)? This can be any member of the team. For data-focused projects, that individual is expected to be key personnel.

Physical samples: blood, urine, feces

Software: https://sdsuswine.shinyapps.io/sows/ web-based interactive version of the AA deposition model

Data collection/management personnel: Crystal Levesque

Provide how the data are organized and managed for all data created and utilized during this project. This could include the relevant data standardization, data formats, data definitions (e.g., data dictionary, ontologies), data organization, and/or where the data is stored. Please note if any of these data are confidential, and if so, when you would anticipate being able to share them publicly in the future.

Data for each trial are contained within a separate file housed electronically on the South Dakota State University cloud-based server. Specific algorithms related to pattern of AA deposition in each pregnancy-related tissue are confidential at this time. Invention disclosure form has been submitted December 2021.

FFAR supports the FAIR principles. We would like to understand how your project aligns with FAIR principles (for definition, see [Go FAIR], for more information see [Nature Magazine]). Does your data align with the 4 principles in FAIR? Please check all boxes that apply and provide a brief description:

Responses Selected:

Findable – how could your data be found (e.g., online searches, publications)?
Access - how will the data be accessed and shared by researchers and the general public, respectively (e.g., is the data publicly accessible, proprietary, temporarily proprietary)?
Reusability - describe the policies and provisions for re-use, re-distribution, and the production of derivatives.

For each checked box, please provide a brief description.

Findable: PI contact information to request data is included in all published manuscripts.

Accessible: raw data are stored in standard excel and word document formats

Reusability: details related to experimental treatment, animal ID, and other key identifiers needed to reuse the data are included in each excel/word file.

What are the provisions for the data for the appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements?

Data are made available upon request to the Principal Investigator. Institutional data security protocols are used for securing stored/archived data and maintaining data integrity.

Describe the plans for archiving data, samples, and other research products, and for preservation of access to them. Refer to the FFAR Data Guidance Document for more information.

Electronic data will be stored/archived on South Dakota State University cloud-based server. SDSU Central Administration has continual access to all stored data. Raw biological samples are stored in the Principal Investigator's laboratory until the specific data set has been published in peer-review journal. Following publication, all raw samples are destroyed according to institutional protocols.

Will this project coordinate with existing FFAR-supported grants and their data products? If so, provide a brief description of how, including the project title and PI of the existing FFAR work If not, provide a brief rationale for why this project does not coordinate with existing FFAR-supported work.

This project does not directly coordinate with an existing FFAR-supported grant or data products; however, this project does contribute to the overall objective of the FFAR-National Pork Board project on Pig Survivability. As one of the outcomes of the practical feeding trial conducted herein, phase feeding during gestation improved offspring robustness as evidenced by improved post-weaning growth. More robust pigs are also pigs with greater survivability.

Please provide any feedback on these new data questions including any areas where you would like to see more clarity or explanation.

Explanations associated with the new data questions are helpful.

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked 'owner' in parentheses on the menu on the left of the screen when you open your award) can hit 'Submit' on finished tasks. If you are not the owner, the 'Submit' button will not appear. FFAR does not receive notification of completed tasks without a submission.

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5. CERTIFICATION

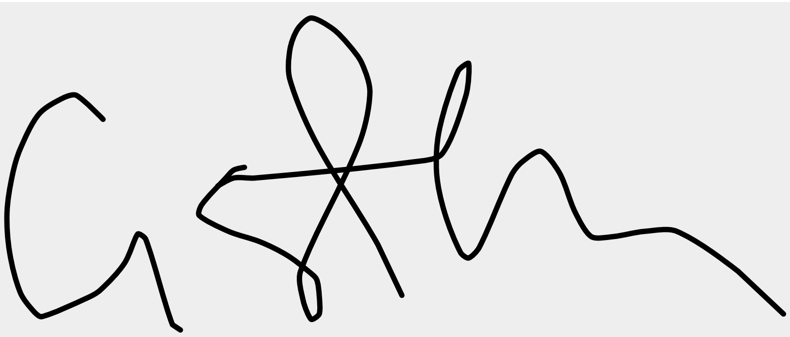
The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information.

By signing below, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name*

Dr. Crystal Levesque

PI Electronic Signature*



Date*

01/20/2022

Authorized Signing Official (ASO)*

Dr. Dianne Nagy

ASO Electronic Signature*

Date*

01/21/2022

Final Invention Report and Certification

Completed - Jan 25 2022

Form for "Final Invention Report and Certification"

FINAL INVENTION REPORT AND CERTIFICATION

A Final Invention Report must be submitted within 90 days after the expiration or termination of a FFAR funded research grant. The Report must include all inventions which were conceived or first actually reduced to practice during work performed under the funded grant, from the original effective start date of the grant through the date of completion or termination. The Final Invention Report and Certification in no way relieve the principal investigator, project director, or the grantee, of the obligation to assure that all inventions are promptly and fully reported directly to the Foundation for Food and Agriculture Research (FFAR), as required by terms of the grant or award.

If no inventions were conceived or reduced to practice during the funded research, write a statement to that effect in the first block under item Title of Invention. The Authorized Organization Representative (AOR) must certify the Report prior to submitting it.

Grantee:*

South Dakota State University

Grant ID:*

0000000004

Project Title:*

Redefining amino acid requirements in pregnancy for precision sow feeding

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Date of the Report:*

12/25/2022

INVENTION(S)

If no inventions were conceived or reduced to practice during the funded research, write “NONE” in the space provided below.

	Name of Inventor	Title of Invention	Date Reported to FFAR
1	Christian Ramirez-Camba	Model of growth, protein, and amino acid deposition in the whole body of the pregnant sow	12/25/2022
2			
3			
4			
5			
6			
7			
8			
9			
10			

The undersigned hereby certifies that to the best of their knowledge and belief, the information in the Report below are true, accurate, and complete. You also certify that the Report lists all inventions which were conceived and/or first actually reduced to practice during the work under the above-referenced FFAR grant or award for the period of:

Project Start Date:*

12/31/2016

Project End Date:*

12/30/2021

Authorized Organization Representative (AOR) Name:*

Dr. Dianne Nagy

AOR Title:*

Research Integrity and Compliance Officer

PI Signature:*

Date:*

12/25/2022

AOR Signature:*

Date:*

01/25/2022

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked ‘owner’ in parentheses on the menu on the left of the screen when you open your award) can hit ‘Submit’ on finished tasks. If you are not the owner, the ‘Submit’ button will not appear. FFAR does not receive notification of completed tasks without a submission.

Final Financial Status Report

Completed - Jan 26 2022

Please download the [Final Financial Status Report] and upload the completed form.

SA1700392 Annual FSR_Year 5

Filename: SA1700392_Annual_FSR_Year_5.pdf **Size:** 122.3 kB

Final Equipment Inventory Report

Completed - Jan 26 2022

Final-Equipment-Inventory-Jan2022-signed

Filename: Final-Equipment-Inventory-Jan2022-signed.pdf **Size:** 679.0 kB

Copies of Relevant Metadata Records

Completed - Jan 31 2022

Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products.

Final Report_Relevant Metadata Report

Filename: Final_Report_Relevant_Metadata_Report.docx **Size:** 12.1 kB

Other Modifications

Incomplete

No Cost Extension Request

Incomplete

Form for "No Cost Extension Request"

NO COST EXTENSION REQUEST

This document describes the procedures that must be followed when a grantee determines that their activities will not be completed during the approved project timeline. A no-cost extension requires prior approval from the Director of Grants Management.

FFAR grants and cooperative agreements are awarded under a project period system. A project may be approved for multiple years (usually 3-5 years), and is generally funded in annual increments known as budget periods. With rare exceptions, budget periods are 12 months in duration. FFAR expects that grantees will complete all requirements of an award by the project period end date; however, a one-time no cost extension may be requested if the action is not prohibited in the grant agreement. The Director of Grants Management can approve up to 6 additional months but no additional funds are awarded to complete the funded tasks.

Note that a No Cost Extension Request should not be submitted for the sole purpose of expending remaining funds – such request will be disapproved. No Cost Extensions cannot be processed under expanded authority if the award project period end date has expired.

FFAR Notification

To ensure timely processing of a revised award action and orderly accomplishment of activities, a no-cost extension should be requested at least 45 days prior to the end of the project period by sending a request on official grantee letterhead that includes the following:

- Date
- Principal Investigator or Project Director name, Project Title, and Grant ID
- Point of contact – name, phone number, and email address
- Amount of additional time requested
- Reason(s) project could not be completed
- Description of the activities that will be completed during the proposed extension
- Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR
- For late requests, a justification for missing the deadline
- Explain the effect a denial of the request will have on the project
- Two signatures – Authorized Signing Official and Project Director/Principal Investigator

NO COST EXTENSION REQUEST

Date:

(No response)

Principal Investigator or Project Director name

(No response)

Project Title

(No response)

Grant ID

(No response)

Point of contact

Name	(No response)
Phone Number	(No response)
Email Address	(No response)

Amount of additional time requested:

(No response)

Reason(s) project could not be completed:

(No response)

Description of the activities that will be completed during the proposed extension:

(No response)

Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR

(No response)

For late requests, a justification for missing the deadline

(No response)

Explain the effect a denial of the request will have on the project

(No response)

Authorized Signing Official

Date Signed:*

(No response)

Project Director/Principal Investigator

Date Signed:*

(No response)

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked ‘owner’ in parentheses on the menu on the left of the screen when you open your award) can hit ‘Submit’ on finished tasks. If you are not the owner, the ‘Submit’ button will not appear. FFAR does not receive notification of completed tasks without a submission.

Optional - Additional Materials Upload

Incomplete

GRANT INFORMATION	
Award Program	New Innovator in Food and Agriculture Research
Grant ID	430840
Project Title	Redefining amino acid requirements in pregnancy for precision sow feeding
Reporting Period	12/01/20 - 12/31/21
PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR INFORMATION	
Full Name	Dr. Crystal Levesque
Email	Crystal.Levesque@sdstate.edu
Phone	605-688-5011
GRANTEE ORGANIZATION INFORMATION	
Organization Name	South Dakota State University
Address	1015 Campanile Ave, SAD 200, Box 2201
City, State Zip Code	Brookings, SD 57007
EIN	46-6000364
AUTHORIZED SIGNING OFFICIAL INFORMATION	
Name	Dianne Nagy
Email	Dianne.Nagy@sdstate.edu
Phone	605-688-5051

Green cells will autofill

REPORT OF RECEIPTS AND EXPENDITURES

REVENUE

Total Approved FFAR Award for the grant	$ 63,379.50
Total Approved Cash Match for the grant	$ 63,379.50
Total Approved In-Kind Match for the grant	
Total Approved Committed Matching Funds	$ 63,379.50
Grant Award Total	$ 126,759.00

Did you request a re-budget (re-allocation of the original budget) for this budget period?	No

EXPENDITURES *(For the entire grant duration. Please refer to your submitted annual report(s), if applicable, for yearly expenditure totals)*

FFAR Funds Expended

Approved Budget Line Item Categories				
Personnel	Budgeted Cost	Actual Cost	Difference	% Variance
Total Salary	$ 31,390.00	$ 1,702.25	$ 29,687.75	-95%
Total Fringe Benefits	$ 2,195.00	$ 13.81	$ 2,181.19	-99%
Total Personnel Expenditure	$ 33,585.00	$ 1,716.06	$ 31,868.94	-194%

Direct Costs	Budgeted Cost	Actual Cost	Difference	% Variance
Equipment (over $5,000)			$ -	#DIV/0!
Travel & Accommodations	$ 1,800.00	$ 1,244.93	$ 555.07	-31%
Conferences, Coventions, Meetings			$ -	#DIV/0!
Materials and Supplies	$ 13,000.00	$ 1,139.34	$ 11,860.66	-91%
Printing and Publication			$ -	#DIV/0!
Computer(s) Fees			$ -	#DIV/0!
Data Management			$ -	#DIV/0!
Training Support Costs			$ -	#DIV/0!
Consulting and Professional Fees			$ -	#DIV/0!
Subaward(s)/Contractual Costs			$ -	#DIV/0!
Other Direct Costs	$ 8,657.00	$ 4,192.98	$ 4,464.02	-52%
Total Direct Costs	$ 57,042.00	$ 8,293.31	$ 48,748.69	#DIV/0!

Indirect Cost (IDC)	Budgeted Cost	Actual Cost	Difference	% Variance
Indirect Costs	$ 6,337.50	$ 921.39	$ 5,416.11	-85%
Total Indirect Costs	$ 6,337.50	$ 921.39	$ 5,416.11	-85%

	Budgeted Cost	Actual Cost	Difference	% Variance
Total Expended FFAR Funds	$ 63,379.50	$ 9,214.70	$ 54,164.80	-85%

Matching Funds Expended

Approved Matching Line Item Categories	**Committed Match**		**Actual Match Expended**		**Difference**		**% Variance**	
Personnel	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind
Total Salary	$ 31,390.00		$ 56,595.97		$ (25,205.97)	$ -	80%	#DIV/0!
Total Fringe Benefits	$ 2,195.00		$ 4,060.39		$ (1,865.39)	$ -	85%	#DIV/0!
Total Personnel Expenditure	$ 33,585.00	$ -	$ 60,656.36	$ -	$ (27,071.36)	$ -	81%	#DIV/0!

Direct Costs	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind
Equipment (over $5,000)					$ -	$ -	#DIV/0!	#DIV/0!
Travel & Accommodations	$ 1,800.00		$ 2,112.55		$ (312.55)	$ -	17%	#DIV/0!
Conferences, Coventions, Meetings					$ -	$ -	#DIV/0!	#DIV/0!
Materials and Supplies	$ 13,000.00		$ 12,587.05		$ 412.95	$ -	-3%	#DIV/0!
Printing and Publication					$ -	$ -	#DIV/0!	#DIV/0!
Computer(s) Fees					$ -	$ -	#DIV/0!	#DIV/0!

Data Management					$ -	$ -	#DIV/0!	#DIV/0!
Training Support Costs					$ -	$ -	#DIV/0!	#DIV/0!
Consulting and Professional Fees					$ -	$ -	#DIV/0!	#DIV/0!
Subaward(s)/Contractual Costs					$ -	$ -	#DIV/0!	#DIV/0!
Other Direct Costs	$ 8,657.00		$ 53,376.52		$ (44,719.52)	$ -	517%	#DIV/0!
Total Direct Costs	$ 57,042.00	$ -	$ 128,732.48	$ -	$ (71,690.48)	$ -	126%	#DIV/0!

Indirect Cost (IDC)	Committed Match	Actual Match Expended	Difference	% Variance
Indirect Costs	$ 6,338.00	$ 14,302.18	$ (7,964.18)	126%
Total Indirect Costs	$ 6,338.00	$ 14,302.18	$ (7,964.18)	126%

Note that no part of the
IDC can be a match

	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind	Cash	In-Kind
Total Matching Funds	$ 63,380.00	$ -	$ 143,034.66	$ -	$ (79,654.66)	$ -	126%	#DIV/0!

	FFAR Funds	**Matching Funds**
ENCUMBRANCES (invoiced or billed but not paid) BALANCE:		

	FFAR Funds	**Matching Funds**
NET EXPENDITURES (actual expenditures + encumbrances):	$ 9,214.70	$ 143,034.66

	FFAR Funds	**Matching Funds**
WE REPORT A BALANCE IN THE AMOUNT OF:	$ 54,164.80	$ (79,655.16)

YEAR TO DATE TOTALS FOR THE GRANT	
Total FFAR Funds Expended	$ 9,214.70
Total Matching Funds Expended	$ 143,034.66
Total Expended Funds	$ 152,249.36
Total Expended Indirect Costs	$ 15,223.57
Total Expended IDC (FFAR Funds)	$ 921.39
Total Expended IDC (Matching Funds)	$ 14,302.18
Total Reporting Balance	$ (25,490.36)

By signing this report, I certify that all expenditures reported or payments requested are for appropriate purposes and are in accordance with the agreement set forth in the award documents. I am aware that any false, fictitious, or fraudulent information, or the omission of any material fact, may subject me to criminal, civil, or administrative penalties for fraud, false statements, false claims or otherwise. By typing my name here, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name	Crystal Levesque	**ASO Full Name**	**Dianne Nagy**
PI Electronic Signature	☑	**ASO Electronic Signature**	☑
Date of Certification		**Date of Certification**	

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Equipment Inventory Report

The Equipment Inventory Report must be submitted as part of the Final Report within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

A complete inventory must be submitted for all major equipment acquired or furnished under this project with a unit acquisition cost of $5,000 or more. The inventory list must include the description of the item, manufacturer serial and/or identification number, acquisition date and cost, percentage of FFAR funds used in the acquisition of the item.

Equipment with a unit acquisition cost of less than $5,000 that is no longer to be used in projects or programs currently or previously funded by the FFAR may be retained, sold, or otherwise disposed of, with no further obligation to FFAR. If no equipment was acquired under this award, please write a statement to that effect.

Grantee Name: South Dakota State University

Project Title: Redefining amino acid requirements in pregnancy for precision sow feeding

Grant ID: 0000000004

Principal Investigator: Crystal Levesque

Grant Period: 12/01/2016-12/31/2021

Report Date: 1/26/22

Item Description	Manufacturer	Serial Number	Quantity	Condition [1]	Location [2]	Purchase Cost	% of FFAR funds to purchase	Date Received
no items to report								

1 Condition: Excellent, Fair, Poor or Inoperable
2 Location: complete physical address.

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information. If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Crystal Levesque

PI Signature: DocuSigned by: Crystal Levesque 19EDFABDED30446... Date: 1/26/22

Authorized Signing Official (ASO): Dianne Nagy

ASO Signature: DocuSigned by: Dianne Nagy 7670CD5E2E48405... Date: 1/26/22